THE AUTONOMY PARADOX: WHEN DOES AGENTIC AI CREATE OR DESTROY ORGANIZATIONAL VALUE?

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Abstract. We model the autonomy of an AI agent as a bundle of decision rights to plan, execute and observe, subject to error cascades within unsupervised runs and a hazard at every hand-off between person and agent. We characterize when partial autonomy is optimal and show that splitting planning from execution can make a little autonomy worse than none. A connection between the hand-off hazard and the value of codifying work is given. In 7,296 benchmark trajectories and 360 conversations with a 2026 agent, the split allocation of rights is the worst design. Control algorithms, a telemetry audit and two deployment workflows follow from the model.

1. Introduction

Firms now let software act, not only advise. An agent in a customer-operations queue no longer drafts a reply for a person to send. It reads the ticket, queries the account, calls a provisioning tool, tells the customer which setting to change on a device it cannot see and closes the case. The first wave of generative AI raised productivity while leaving a person to accept or reject each output [7, 9]. Agentic systems aim to remove most of that checkpoint. Their failures seldom look like a single wrong answer. They look like a run that drifts, in which every later tool call builds on an early misreading, or an instruction that is clear when the agent issues it and lost by the time a person acts on it. For a general manager this is an old question of organization design in a new form: which decisions this actor may take, who must see the work before it moves on and who answers when it fails.

Practice answers it incrementally: start with a person approving most steps, then widen the agent's remit as its measured accuracy improves. This answer assumes that value rises with autonomy, so that capability sets the right level of autonomy. Existing theory was not built to test it. IS scholarship frames the use of agentic artifacts as delegation [4] and identifies autonomy as a defining facet of managing AI [6], yet it treats the delegated unit as a task or a role. The human–AI collaboration literature explains failures of joint work mainly through human limitations such as aversion, overreliance and complacency [10, 12, 25]. The automation–augmentation debate leaves open when a mixed allocation beats either pole rather than serving as a waystation between them [29]. The evidence does not fit the monotone view either. In a controlled benchmark, an agent that diagnoses a problem and tells a simulated customer what to do performs worse than the agent alone and worse than the customer working from a correct script.

We argue that "autonomy" is the wrong unit of analysis. In an agentic workflow it is a bundle of decision rights over a sequence of steps: the right to *plan* what should be done, the right to *execute* each action and, bound up with execution, the right to *observe* the state of the work. The value of granting one right depends on who holds the others. Decisions belong where the knowledge they require resides [16]. Formal authority without the information to exercise it confers little real authority [1]. An agent that plans a repair on a phone it cannot observe holds formal authority over the case but little real authority over its outcome. What

Key words and phrases. Agentic AI, decision rights, delegation, human oversight, organizational control, AgentOps, error cascades, human–AI collaboration.
Manuscript prepared for *Information Systems Research*.

agentic AI adds is the ability to cut one workflow into steps whose rights belong to different kinds of actors. Each cut is a coordination interface with costs of its own [22, 33].

Two mechanisms set those costs and pull in opposite directions. The first is the *error cascade*. An agent acting for many steps without a human touchpoint commits to its earlier reading of the case and carries any early misdiagnosis forward. Its per-step hazard therefore rises with the length of the unsupervised run, much as failures propagate through tightly coupled systems [27]. Touchpoints interrupt such runs. The second is the *boundary-crossing hazard*. A decision can be misread each time it passes between agent and human, as an instruction the person must carry out or as a state the agent learns about second-hand. The hazard belongs to the interface. The organization pays it on every step that the agent plans and a person executes. Cascades argue for more touchpoints; boundary crossings argue for fewer. We specify both mechanisms inside one model of delegated work, so their trade-off can be derived and estimated.

The model yields four results. First, without cascades, partial autonomy is never optimal: value is convex in the number of delegated steps, so the optimal autonomy level lies in $\{0, 1/n, 1\}$. Second, with cascades, the most valuable touchpoint is the last one; its value grows with the square of workflow length. A firm can optimally deny full autonomy over long workflows to an agent that is more reliable than a person at every step. Third, when the agent plans and the human executes, the organization can enter a *valley*. There a little autonomy is worse than none and, in some regimes, worse than full autonomy. The reason is that the first delegated decision moves every remaining human step across the boundary. Fourth, reviewer vigilance and agent observability are substitutes. They affect value only through an effective cascade intensity, $\rho(2-\kappa)/\kappa$. Halving vigilance therefore triples the effective intensity. Better tracing and checkpointing can buy back that loss.

We test the theory in eight studies. Their core is 7,296 trajectories from τ^2-bench [5]. In them two agents (GPT-4.1 and o4-mini) ran the same 114 telecom tasks under four allocations of planning and execution rights, two policies and four trials. Further studies use the public leaderboard and pass^k [35], a 2026 agent, Collaborative Gym [31], real users [31, 32] and multi-agent failure logs [8, 37]. Shared control was the worst design in all four agent–policy cells, 23 to 57 percentage points below the best bundled design. Delegating execution raised success by 18.0 points when the agent also planned but by only 5.6 points when the plan was specified (interaction −12.3, SE 2.8). The data reject $\rho = 0$ in 6 of 8 agent–policy–design cells. With fault-count fixed effects, however, the curvature survives only for o4-mini; for the other agents we cannot separate cascades from diagnostic difficulty. Codifying the policy as an explicit workflow cut the boundary-crossing hazard by a median 76% for GPT-4.1 and 58% for o4-mini. On indicative leaderboard estimates, the shared-control hazard fell from a median 0.26 to 0.025 between June 2025 and 2026.

The newer evidence is narrower and mixed. In 360 new conversations with a 2026 agent (GPT-5.6 Luna, 45 tasks per design under each policy), shared control was again the worst design. Under the prose policy it succeeded in 35.6% of tasks, against 60.0% for full autonomy and 80.0% for the human-led design; the pre-specified valley test held. Under the codified policy it trailed full autonomy by only 8.9 points ($p = 0.12$), so the test did not hold. It still trailed the human-led design, as Corollary 1 implies (a check not pre-registered). For this agent the substitution interaction was not negative (+5.6 points, SE 9.4). Its cascade estimates ($\hat{\rho} \approx 0.05$) exceeded those of 2025 but are again confounded with difficulty. In ten paired data-analysis tasks where a simulated user held knowledge the agent lacked, collaboration did not raise scores (a mean rating of 0.80 in both conditions). Among real users, rated outcomes peaked at an intermediate human share (about 0.30), the opposite of the pre-specified valley. We read this as a possible boundary of the theory, which our other evidence does not confirm. In orchestrator-led multi-agent runs, decisive errors fell on hand-off steps 1.69 times as often as chance ($p < 0.001$); in expert teams they did not (0.98).

The paper makes three contributions. The first is to IS theories of delegation and algorithmic control [4, 6, 18]. Our model takes a decision right within a workflow, not a task or a role, as the unit of delegation. In it the value of delegating one right depends on how the others are allocated. Human touchpoints become behavior controls that people exercise over machine actors [19, 24]. The second is to research on human–AI collaboration [2, 12, 29]. We identify two failure mechanisms that operate even with competent people and call for different remedies: touchpoints, checkpoints and observability for cascades; codification and bundled rights for boundary crossings. We also state when augmentation is optimal rather than transitional. The third is to practice and AgentOps. We provide a telemetry audit of the model's parameters and an exact dynamic program for placing touchpoints. We also give a gate that moves the planning right on the evidence from boundary crossings, an in-run checkpointing rule that measures reviewer vigilance and two deployment workflows. The design rules are to keep rights bundled, bound unsupervised runs, codify before delegating and treat observability and accountability as substitutes priced at a known exchange rate. Section 2 reviews the background and Section 3 develops the model. Sections 4 and 5 present the design and studies, Section 6 the procedures for practice and Section 7 the implications and limitations. Section 8 concludes.

2. Theoretical Background

The model in Section 3 draws on organizational research on decision rights, delegation and control, on the reliability and human–AI collaboration literatures and on technical work on how language-model agents act and fail. We ask of each what it explains and where it stops once the actor receiving decision rights is an agent inside a multi-step workflow.

2.1. **Decision rights and delegation.** Organizational economics holds that decisions are best taken where the knowledge needed to take them resides. When that knowledge is costly to transfer, rights should move to the knowledge. The organization must then build systems to control those who hold them [16]. Aghion and Tirole [1] separate formal authority, the right to decide, from real authority, effective control over the decision. The two diverge when the holder of the formal right lacks the information to use it. In multitask agency, the way tasks are grouped into jobs shapes the cost of controlling the people who do them, so the value of assigning one task depends on what else the job contains [15]. Our claim that the value of one right depends on who holds the others has the same structure, applied to the steps of a workflow and operating through information and error rather than incentives. Organization design adds the costs of dividing interdependent work. Thompson [33] matches sequential and reciprocal interdependence to coordination by plan and by mutual adjustment, each costlier than coordination by standardization. Malone and Crowston [22] define coordination as the management of dependencies. Galbraith [13] frames design as matching information-processing capacity to task uncertainty and Puranam et al. [28] reduce organizing to dividing labor, allocating tasks, providing information and rewarding effort.

IS research supplies the lens of delegation. Baird and Maruping [4] reframe IS use around agentic artifacts that take on rights and responsibilities from humans, who take some back in an ongoing exchange; their framework is the starting point of our model. Berente et al. [6] identify autonomy, learning and inscrutability as what makes AI hard to manage. Murray et al. [23] theorize conjoined agency whose forms differ by whether technology helps develop the protocols that govern action, select the actions or both, which anticipates our separation of rights within one course of action. Field research shows organizations rebuilding work around people who audit and alter an algorithm's output [14]. Research on algorithmic control documents how algorithms direct and discipline workers [18]. Control theory ties the choice between behavior and outcome control to knowledge of the process and measurability of outputs [19, 24]. In agentic workflows the direction of control reverses: a human touchpoint, a step at which a person sees the state of the work before it moves on, is a behavior control

exercised over a machine actor. Its worth is limited by what the person can see. For this reason observation travels with execution in our model.

These traditions explain why rights should sit with knowledge and why splitting work is costly. Yet they assume that every actor is a person and that jobs or tasks group the rights. Agentic AI breaks both assumptions. The knowledge a case requires is now split by kind of actor: a person sees the customer's device or an informal norm, the agent sees the records, the policy text and the tool outputs. The literature says such cuts are costly. It does not say how the cost scales with the number of steps executed across the cut, how it trades off against leaving a machine actor unsupervised or where in a sequence a behavior control should be placed. The boundary-crossing hazard ε formalizes the first cost and the cascade intensity ρ the second. Thompson's categories also locate the two remedies we study: a codified-workflow policy is coordination by standardization, whereas the specified plans of the human-led and executor designs are coordination by plan.

2.2. **Reliability, oversight and human–AI collaboration.** Combining people and AI can raise performance or lower it. Generative AI raised productivity in customer support, most among less experienced workers [7]. It raised consultants' quality inside the technology's capability frontier while lowering it outside [9]. AI advice can improve individual decisions while eroding distinctive human knowledge [12]. Professionals set aside opaque AI they cannot interrogate [20]. People abandon algorithms after seeing them err [10], so human-factors research calls for trust calibrated to capability [21]. Athey et al. [2] model the allocation of authority between a person and an AI as a trade-off between AI accuracy and human effort. Raisch and Krakowski [29] argue that automation and augmentation are interdependent over time. This literature locates poor joint performance mainly in the person and studies one decision at a time, so it cannot represent faults that compound across steps.

Interactive complexity with tight coupling produces failures that spread faster than operators can stop them [27]. Accidents arise when latent errors combine with active errors or local triggers [30]. High-reliability organizations survive through sensitivity to operations [34]. An agent's early misreading behaves like a latent error that lies in its context until later steps act on it. Overseers lapse too: automation removes people from the routine work that keeps them skilled [3] and breeds complacency and automation bias [25]. The types-and-levels framework of Parasuraman et al. [26] is the closest antecedent to our bundle of rights: information acquisition and analysis correspond to observation; decision and action selection to planning; action implementation to execution. It sets a level for each stage but does not model what is lost when adjacent stages are held by different actors, nor how errors compound across a long run held by one actor.

Our model adds two mechanisms that operate even when people are competent and calibrated. Cascades are a property of unsupervised runs; boundary crossings are a property of the interface between planner and executor. The model thereby gives the automation–augmentation debate a condition rather than a tendency. An allocation strictly between $1/n$ and full autonomy, in which humans and agents both hold a substantial share of execution, is optimal only when cascades exist and touchpoints are cheap enough to contain them. The vigilance parameter κ, the probability that a touchpoint is a genuine inspection rather than a rubber stamp, enters value only together with ρ. The complacency warning thus becomes an exchange rate between accountability and observability.

2.3. **Agentic AI as a technical artifact.** A language-model agent acts in a loop: it reads its context, reasons, selects a tool call or message and appends what it observes [36]. Each step therefore conditions on the agent's own earlier outputs. An early misreading becomes evidence for every later step. Transformer accuracy falls sharply as composed steps grow and early errors compound [11]. Those tasks, however, are not tool use in deployed workflows. We therefore treat cascades as a mechanism to be tested: we estimate ρ and allow the estimate to be zero. The agent also observes the world only through its tools. When a customer must

act on a device the agent cannot reach, the agent learns the result only from the customer's report, which is the micro-foundation of the boundary-crossing hazard.

τ-bench places an agent between domain tools and a simulated user, scores the final state of the environment and introduces the pass^k reliability metric [35]. τ^2-bench adds dual control, in which the user also acts on a shared world, with ablations that remove the user or give the agent the ground-truth plan [5]. Read as organization design, those ablations reallocate planning and execution rights with the task fixed; Studies 1–3 use them for that purpose. Evaluations should also report cost [17]. Collaborative Gym lets an agent and a human both act on one task [31]. CollabSkill records real workers completing professional tasks with agents [32]. Who&When attributes failed multi-agent runs to an agent and step [37]. MAST groups fourteen failure modes into specification failures, inter-agent misalignment and verification failures [8]. In our terms these correspond to planning failure ϕ, a boundary-crossing hazard between machine actors and touchpoints that fail to inspect. Studies 6–8 use these resources. They were built to evaluate agents, not to design organizations. A simulated user is not a customer. We use them to estimate the parameters of an organizational model and treat the simulated settings as a laboratory rather than a field site.

Three questions remain open across these literatures: when partial autonomy is optimal, why some divisions of labor perform worse than either party alone and at what rate codification, accountability and observability substitute for one another. The model in Section 3 addresses all three. It treats autonomy as a bundle of rights to plan and to execute, allocated step by step with observation bound to execution.

3. A Decision-Rights Theory of Agentic Workflows

The model takes the view of a manager who owns a recurring workflow and makes two decisions: who determines the steps of a case and who carries out each step. The parameters are properties of the work, the agent and the organization's practices. Sections 4 and 5 estimate most of them and Section 6 shows how a firm can audit the rest.

3.1. **Primitives.**

A workflow consists of n steps, all of which must succeed for the case to succeed. The manager allocates two kinds of rights. The *planning right* $\pi \in \{H, A\}$ determines whether the human or the agent decides what the steps should be: the right to diagnose the case and write its runbook. The *execution right* for each step i belongs to $x_i \in \{H, A\}$. Let $k = |\{i : x_i = A\}|$ be the number of agent-executed steps and $a = k/n$ the *autonomy level*. Observation travels with execution: the executor of a step also observes its result. When a person acts on a device the agent cannot reach, the agent learns the result only from the person's report.

Four quantities govern success.

- *Planning failure* ϕ_π: the probability that the plan itself is wrong (a misdiagnosis).
- *Execution hazard* p_H for a human step and $p_A + \rho u_i$ for an agent step, where u_i is the number of consecutive agent steps since the last human touchpoint. The parameter $\rho \geq 0$ is the *cascade intensity*: how quickly an unsupervised run degrades.
- *Boundary-crossing hazard* $\varepsilon \geq 0$: the extra failure probability of a step that a person executes on the agent's plan, because the decision must cross the human–agent interface. The baseline model sets the reverse direction, an agent executing a specified plan, to zero.
- *Touchpoints*: every human-executed step is a moment at which a person sees the state of the work and resets the run counter u.

Writing h_i for the hazard of step i, the probability that the case succeeds is

$$S = (1 - \phi_\pi) \prod_{i=1}^{n} (1 - h_i), \qquad h_i = \begin{cases} p_A + \rho u_i & x_i = A \\ p_H + \varepsilon \, \mathbf{1}[\pi = A] & x_i = H, \end{cases} \tag{1}$$

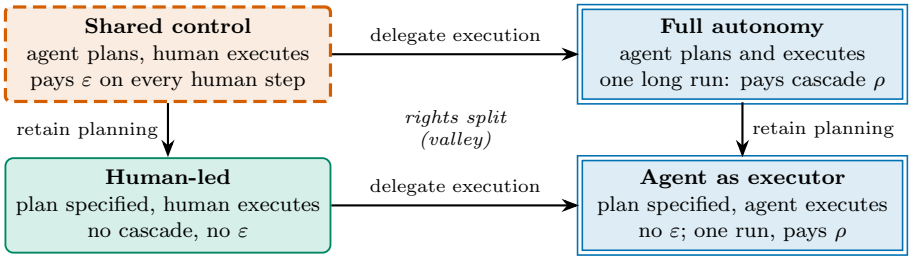


FIGURE 1. Four allocations of planning and execution rights. Arrows show the two delegation moves. The theory predicts that the design in which a person executes the agent's plan (top left) is dominated. It also predicts that the value of delegating execution is larger when the agent holds planning (top arrow) than when planning is specified (bottom arrow). Shared control's dashed border marks the split. Color online; in print, shapes and line styles carry the same coding.

and the organization's value per case is

$$V = R \cdot S - c_H\,(n-k) - c_A\,k, \tag{2}$$

where R is the value of a correctly resolved case and $c_H > c_A$ are the per-step costs of human and agent effort. Equivalently, $V = R - [c_H(n-k) + c_A k] - R(1-S)$: every design choice acts on risk through S and on operating cost through k.

Each primitive is an organizational fact. The cascade intensity ρ asks how long the agent can run unattended. We estimate it, allowing it to be zero. The boundary-crossing hazard ε asks how costly it is to make one party act on another's plan. Because observation travels with execution, the allocation of the x_i determines who holds the information that confers real authority [1]. Each touchpoint is a behavior control exercised over a machine actor [24]. An agent executing a specified plan observes its own results, so only the instruction half of the interface remains. As a simplification, the baseline sets it to zero.

3.2. **The autonomy path and the four corner designs.** On the *autonomy path*, at $k = 0$ the human plans and executes everything: a human-led workflow with no boundary crossings and no agent runs. For every $k \geq 1$ the agent holds the planning right and executes k steps. The remaining $n-k$ human steps serve as touchpoints, each crossing the boundary. At $k = n$ the agent plans and executes alone. The path is discontinuous at its start: the first delegated step also changes who plans all of them.

Shared control (the agent plans, the human executes most steps) is a point near the start of the path. *Agent as executor* ($\pi = H$, $k = n$) lies off the path: it bundles a human-held plan (in practice, a runbook or a codified procedure) with agent execution in a single run. Figure 1 arranges the four corner designs by the two delegation moves. Three are *bundled*: one party plans and executes, or the agent executes a specified plan and observes its own results. Shared control alone splits planning from execution in the direction that pays ε.

3.3. **Results.** We now state the results of the model. The proofs are in Online Appendix A. We start with a placement lemma, which makes the results tractable.

Lemma 1 (Balanced touchpoints)**.** *For a fixed number k of agent steps and homogeneous hazards, success is maximized by spacing the $n-k$ human touchpoints so that the unsupervised runs between them differ in length by at most one.*

Because cascade losses grow faster than linearly in run length, a fixed budget of review points belongs at regular intervals through a run rather than bunched together. (Touchpoints act on cascades, not on ϕ, so the model gives no credit to a check that catches a misdiagnosis.) We use balanced placement throughout and relax homogeneity in Algorithm S1 (Online Appendix M).

Proposition 1 (No containment, no interior autonomy)**.** *If $\rho = 0$, value along the agent-led part of the autonomy path is convex in k. The optimal autonomy level is therefore $a^* \in \{0, 1/n, 1\}$; no autonomy level strictly between $1/n$ and 1 is ever optimal.*

Supervised autonomy is therefore not a natural middle ground. Without compounding, each delegated step multiplies success by the same factor; an exponential plus a linear cost has no interior maximum. The practical test is whether an agent's failure rate rises with the length of its unsupervised runs. If not, approving some fraction of its actions is never better than the best of $a = 0$, $a = 1/n$ and $a = 1$. In 4,000 randomly drawn parameterizations with $\rho = 0$, the numerical optimum was never interior. With $\rho > 0$ it was interior in 50% of draws.

Proposition 2 (The last touchpoint and the horizon threshold)**.** *Assume $\rho > 0$ and balanced runs, and let $\delta = p_H + \varepsilon - p_A$ be the per-step reliability advantage of the agent. Replacing the last human touchpoint by an agent step changes log-success by approximately*

$$\Delta \log S \;\approx\; \delta \;-\; \rho\,\frac{(n-1)(n+3)}{4}. \tag{3}$$

Full autonomy is preferred to its nearest supervised alternative only for workflows shorter than

$$\bar{n} \;\approx\; 2\sqrt{(\delta+\gamma)/\rho}, \qquad \gamma = -\ln\!\left(1 - \frac{c_H - c_A}{R\,S_{n-1}}\right). \tag{4}$$

This is the autonomy paradox. The last touchpoint is the most valuable one, because removing it joins two runs into the longest run. Its value rises with the square of workflow length. Beyond $\bar{n}$ it can be optimal to deny full autonomy even to an agent that is more reliable than a person at every step ($\delta > 0$). Capability raises the threshold only with the square root of the reliability gap. Cascades lower it and the boundary-crossing hazard raises it, because error-prone touchpoints make oversight expensive. Approving full autonomy for short cases therefore does not license it for long ones.

Proposition 3 (The unbundling valley)**.** *Moving from the human-led design ($k = 0$) to the first unit of autonomy ($k = 1$) multiplies success by*

$$\frac{S_1}{S_0} = \frac{1-\phi_A}{1-\phi_H} \cdot \frac{1-p_A}{1-p_H} \cdot \left(1 - \frac{\varepsilon}{1-p_H}\right)^{n-1}. \tag{5}$$

For $\varepsilon > 0$ this ratio falls geometrically in workflow length, so the first unit of autonomy destroys a growing share of the human-led design's expected benefit $R\,S_0$. It lowers value, $V(1) < V(0)$, whenever that loss exceeds the labor it saves, $R(S_0 - S_1) > c_H - c_A$: the value of autonomy then falls before it can rise. Moreover, the gain from delegating execution is larger when the agent holds the planning right than when planning is specified; planning retention and execution delegation are substitutes.

Organizations that introduce agents as "assistants" that tell people what to do may thus perform worse than their pre-AI process and, in the U-shaped regimes of the phase diagram, worse than full automation. The first delegated decision moves the whole workflow across the interface. In the terms of Aghion and Tirole [1], the agent acquires formal authority over a case whose state it cannot observe, while the person who can observe it acts on a diagnosis that is not theirs. The remedy is bundled rights: give the agent the tools to execute what it plans, or specify the plan so that the person executes what the person understands. Delegating execution is most valuable as a repair for a split allocation. Whether the dip occurs at all depends on ε relative to a critical value. Corollary 1 gives the threshold and the effect of codification.

Corollary 1 (Critical boundary-crossing hazard)**.** *Let $S_0 = (1-\phi_H)(1-p_H)^n$ be the success probability of the human-led design. Assume homogeneous steps, $n \geq 2$ and $\Delta c = (c_H -$*

$c_A)/R < S_0$. *Let* $a = \ln \frac{(1-\phi_A)(1-p_A)}{(1-\phi_H)(1-p_H)}$ *and* $\gamma_0 = -\ln(1 - \Delta c/S_0)$. *Then* $V(1) < V(0)$ *if and only if* $\varepsilon > \varepsilon^*$, *where*

$$\varepsilon^* = (1 - p_H)\left[1 - e^{-(a+\gamma_0)/(n-1)}\right] \approx (1 - p_H)\,\frac{a + \gamma_0}{n - 1}. \tag{6}$$

If $a + \gamma_0 \leq 0$, *the dip occurs at every* $\varepsilon > 0$. *If codification scales a hazard* $\varepsilon > 0$ *to* $(1 - c)\varepsilon$, *the depth* $V(0) - V(1)$ *falls strictly in* c; *the dip disappears if and only if* $c \geq 1 - \varepsilon^*/\varepsilon$.

The valley is thus a threshold phenomenon in ε. Lowering the hazard makes it shallower but removes it only once ε falls below ε^*. That threshold is small when the agent's plan and first step are little more reliable than the person's and labor savings are modest, or when the workflow is long. The results so far also assume that every touchpoint inspects the work. The next proposition relaxes this.

Proposition 4 (Vigilance, accountability and observability)**.** *Suppose each touchpoint is vigilant (it inspects, repairs latent faults and resets the run) with probability* $\kappa \in (0, 1]$, *and is otherwise a rubber stamp. With balanced runs of length* m *and* n_A *agent steps, the expected cascade loss is approximately*

$$\rho\, n_A\, \frac{m(2 - \kappa)/\kappa - 1}{2}. \tag{7}$$

Vigilance and cascade intensity therefore enter value only through the effective intensity $\tilde{\rho} = \rho(2 - \kappa)/\kappa$, *and the optimal autonomy level depends on them only through* $\tilde{\rho}$.

A rubber-stamp review costs as much as a genuine one but lets the runs on either side of it merge. Halving vigilance from $\kappa = 1$ to $\kappa = 0.5$ therefore triples the effective cascade intensity. Accountability that keeps reviewers attentive raises κ; AgentOps investment in step-level tracing, checkpoints and rollback lowers $\tilde{\rho}$. The two are substitutes at a known exchange rate. Monte Carlo simulation of the full stochastic process confirms equation (7) within 1.3% at $\kappa = 0.8$ and within 4% at $\kappa = 0.6$. For low vigilance and short workflows the formula overstates the loss, because the end of the workflow truncates merged runs (Online Appendix B).

3.4. **The phase diagram.** Figure 2 maps the optimal design for a 12-step workflow as ε and ρ vary, holding other parameters at reference values. With low ε and low ρ, full autonomy is optimal. With low ε and high ρ, supervised autonomy (an interior a^* with periodic touchpoints) is optimal. With high ε and non-trivial ρ, the human-led design is optimal. At these reference values the first unit of autonomy already lowers value for almost any positive ε (Proposition 3; Corollary 1). To the right of the dashed line, the dip is deep enough that some interior design falls below both the human-led and the fully autonomous design. After that dip, value may rise steadily, trace an inverted U or trace a deep U. The two hazards, not capability alone, determine which shape obtains (Figure S1 in the Online Appendix plots the curves).

The diagram is also a map of levers. Codification moves a workflow left by lowering ε; specifying the plan removes ε altogether. Tracing, checkpointing and accountable review move it down by lowering $\tilde{\rho}$. A model upgrade can move it along both axes, so the firm must re-estimate its position. Online Appendix E discusses the model's boundary conditions.

3.5. **Hypotheses.** The empirical setting observes the four corner designs directly. Hypothesis 1 combines Proposition 3 with the U-shaped regimes of the phase diagram. Hypothesis 5 is the observable form of the proposition's second part. Hypotheses 2 and 3 test the two primitives. A positive ε drives the valley; a positive ρ lifts the corner result of Proposition 1 and gives Proposition 2 its force. Hypothesis 4 tests ε as a property of the interface, which coordination by standardization should reduce. Corollary 1 adds no pre-specified hypothesis. It qualifies Hypotheses 1 and 4: codification can make the valley against the human-led design shallower without removing it. Section 5 checks this reading, which was not pre-registered.

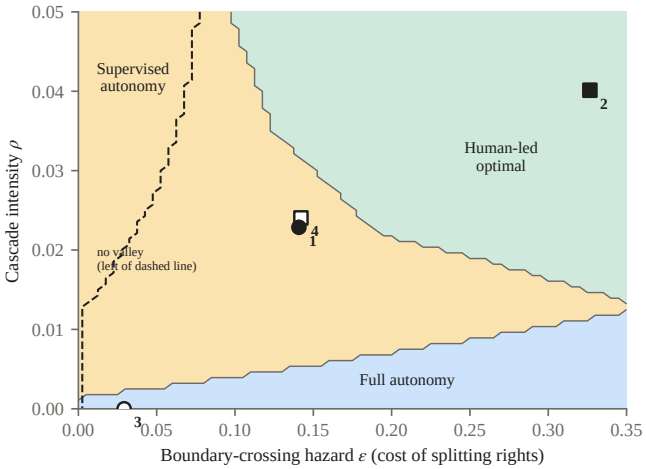


FIGURE 2. Optimal design by boundary-crossing hazard ε and cascade intensity ρ ($n = 12$; $p_A = 0.02$, $p_H = 0.05$, $\phi_A = 0.02$, $\phi_H = 0$, $c_H = 0.02R$, $c_A = 0.001R$). Right of the dashed line, some interior design is worth less than both the human-led and the fully autonomous design. Markers are the calibrated positions of the four model–policy cells from Study 2: (1) GPT-4.1, prose; (2) o4-mini, prose; (3) GPT-4.1, codified; (4) o4-mini, codified. Filled markers are prose policies, open markers codified workflows.

The setting does not vary reviewer vigilance, so we do not test Proposition 4. Section 6 shows how a firm can estimate κ.

Hypothesis 1 (Unbundling valley)**.** Shared control, in which the agent plans and the human executes, yields lower success than both full autonomy and the human-led design.

Hypothesis 2 (Boundary-crossing hazard)**.** Under shared control, success declines geometrically in the number of actions held by the human, at a per-action hazard that exceeds the hazard of the same actions when the plan is specified.

Hypothesis 3 (Cascades)**.** In unsupervised agent runs, success declines faster in workflow length than independent per-step errors imply ($\rho > 0$).

Hypothesis 4 (Codification)**.** Codifying the policy as an explicit workflow reduces the boundary-crossing hazard.

Hypothesis 5 (Substitution)**.** The success gain from delegating execution is larger when the agent holds the planning right than when planning is specified.

4. Research Design

We now describe how the eight studies test the hypotheses, what each one measures and how we estimate the model's parameters.

4.1. **Research strategy.** The theory turns on two parameters, ε and ρ, that no firm records. A credible test needs allocations the actors did not choose, the same work under each allocation and outcomes that do not depend on reading the transcript. Field data rarely satisfy all three. We therefore start in an organizational laboratory that does. We then relax its limits one at a time, with evidence weaker on identification but closer to practice.

Table 1 summarizes the design. Studies 1–3 use a fully crossed experiment on telecom tasks to estimate the effect of each right, recover the two hazards and calibrate the model. Studies 4 and 5 answer the objection that the agents are old. Study 6 tests a boundary condition: work

TABLE 1. Studies, data and the claims each addresses

#	Data	Design	Sample	Tests
1	τ^2-bench telecom logs (2025 agents)	4 designs × 2 agents × 2 policies, within task	7,296 trajectories	H1, H5
2	same	structural hazards by cell	7,296 trajectories	H2, H3, H4
3	Study 2 estimates	calibration of the model	4 agent–policy cells	Props. 1–3
4	public leaderboard	implied hazard; pass^k	21 telecom; 18 banking submissions	drift in parameters (indicative)
5	new τ^2-bench runs, GPT-5.6 Luna	4 designs × 2 policies, within task, 1 trial	45 tasks per design and policy (360 conversations); 8-task pilot	H1–H5
6	new Collaborative Gym runs	autonomous vs. collaborative agent, paired	10 task pairs; 2-pair pilot	boundary condition
7	Co-Gym real users; CollabSkill	observational division of labor	228; 386 sessions	H1 analogue
8	Who&When; MAST/MAD	failure attribution and labels	184 runs; 1,642 traces	H2, H3 analogues

Studies 5 and 6 ran under the spending caps of Section 4.6; their results come from the main phase, with the pilot as provenance only. Studies 7 and 8 use released logs and make no model calls. Banking submissions are those evaluated May–August 2026.

in which the human holds knowledge the agent lacks. Study 7 brings in real people at the cost of experimental control. Study 8 moves the boundary between agents.

4.2. **Setting and design of Studies 1–3: a dual-control laboratory.** We use the trajectory logs released with τ^2-bench [5], taken from the benchmark's public repository at the commit of 24 October 2026. In the telecom domain an agent acts as a customer-service representative for a mobile operator. A language model simulating a customer reports a problem with their phone. Each task combines two to nine underlying faults (airplane mode on, a locked SIM, data roaming off while abroad, an exhausted data allowance), so resolving it takes one to twelve actions. Some actions are possible only on the customer's device (toggling a setting, rebooting) and others only in the operator's back end (refueling data, resuming a line).

The benchmark's designers ran four configurations that map one to one onto the designs of Figure 1:

Shared control: (default mode): the agent diagnoses and plans. It must instruct the simulated customer to perform device actions, which the customer executes and reports on. The agent cannot see the device.

Full autonomy: (no-user mode): the agent holds all tools (device tools included) and resolves the case alone.

Human-led: (oracle-plan mode): the benchmark supplies the correct sequence of actions to the agent, which relays each device action to the simulated customer, who executes it.

Agent as executor: (oracle plan, no user): the agent receives the correct sequence and executes everything.

Each configuration ran with two agents, GPT-4.1 (temperature zero) and o4-mini (high reasoning effort). Each also ran under two versions of the domain policy: the original prose and a "workflow" version that states the same rules as an explicit decision procedure. The simulated customer is GPT-4.1 at temperature zero and follows a scripted scenario. About two-thirds of the scenarios also assign an "easy" or "hard" persona. Each of the 114 tasks ran four times per cell, with a step limit of 200 and the same four random seeds in every configuration: $4 \times 2 \times 2 \times 114 \times 4 = 7{,}296$ trajectories.

Two features of the mapping matter. First, "plan specified" removes the agent's diagnosis, not the relay. In the human-led design the customer still acts on instructions the agent passes on, so that loss is part of the human-step hazard. Second, the specified plan is the benchmark's ground-truth action sequence, the best case of human-held planning. It sets those designs' planning failure close to zero, so their results are an upper bound for a human-held plan (Section 7.4).

4.3. **Identification.** Every comparison in Studies 1–3 is within task. The task, the customer's scenario, the environment, the agent model and the policy text are identical across configurations; only the allocation of rights (with the tools and information it implies) changes. Identification also rests on a customer who behaves the same way whoever holds the plan, which no real customer can guarantee.

In shared control and the human-led design the customer performs the same device actions; only the source of the plan differs. The difference between the per-action hazards of customer-held steps under the two designs is $\hat{\varepsilon}$. It is identified as a slope in the number of customer-held actions, separately from planning failure ϕ, which is an intercept. Because the human-led design still involves a relay, $\hat{\varepsilon}$ is a lower bound on the total cost of the interface.

Identification of the cascade intensity is weaker. In the unsupervised designs the agent executes a single run of n actions, so curvature identifies ρ. Longer telecom tasks combine more faults (task length and number of faults correlate at 0.84), so convexity could reflect diagnostic difficulty rather than compounding errors. The two unsupervised designs bracket the mechanism. Under full autonomy, $\hat{\rho}$ may absorb rising difficulty and overstate compounding. Under the executor design the supplied plan removes the harder diagnosis along with any compounding, so its $\hat{\rho}$ may understate compounding. We report both. In checks we did not pre-register, we add fault-count and persona controls (Online Appendix I).

Trials are not independent. With the simulated customer and GPT-4.1 at temperature zero, repeated runs of a task differ only through the trial seed and residual non-determinism in model serving. o4-mini varies more but conditions on the same scenario. We therefore cluster standard errors by task and resample whole tasks in the bootstrap. The effective sample is thus 114 tasks, not 7,296 runs. For the same reason pass^k exceeds (pass^1)^k whenever outcomes are correlated across attempts or success rates vary across tasks.

4.4. **Measures.** The primary outcome is task success, the benchmark's binary reward. The benchmark computes it from assertions on the final state of the environment and, for 20 of the 114 tasks, on whether required actions were taken. It never relies on a judgment of the transcript. Secondary outcomes are compute cost (agent plus simulated-customer API spend), elapsed simulation time and the number of messages. Cost overstates the compute cost of designs with a customer and omits what a real customer's time would cost. Elapsed time is the simulation's wall-clock duration, not human working time. For each task we record the customer persona and the number of required actions held by the customer (n_u) and by the agent (n_a). Table 2 reports cell means.

4.5. **Estimation.** *Study 1* estimates the effect of each right with a linear probability model with task × agent × policy fixed effects (456 cells) and standard errors clustered by task:

(8)
$$Y_{tmpc} = \beta_E \,\mathrm{ExecAgent}_c + \beta_P \,\mathrm{PlanSpecified}_c + \beta_{EP} \,\mathrm{ExecAgent}_c \times \mathrm{PlanSpecified}_c + \alpha_{tmp} + e_{tmpc}.$$

With shared control omitted, β_E is the value of delegating execution when the agent plans, $\beta_E + \beta_{EP}$ its value under a specified plan and $\beta_{EP} < 0$ is Hypothesis 5. Because these fixed effects absorb persona, a task attribute, we estimate its interaction with design with agent × policy fixed effects and a control for task length.

Study 2 estimates the structural parameters of equation (1) by maximum likelihood with a Bernoulli likelihood, separately for each agent, policy and design. For designs with human touchpoints, $-\log S = \phi + h_u n_u + h_a n_a$, with per-action hazards h_u and h_a. For unsupervised

TABLE 2. Outcomes by decision-rights design (means over 456 trajectories per cell)

Policy	Design	Success		Cost ($)	Minutes
		GPT-4.1	o4-mini	mean	mean
Prose	Shared control	0.342	0.421	0.181	2.58
	Full autonomy	0.515	0.673	0.078	1.31
	Human-led	0.728	0.963	0.100	1.16
	Agent as executor	0.721	0.989	0.072	0.97
Codified	Shared control	0.518	0.594	0.154	1.71
	Full autonomy	0.684	0.721	0.084	1.41
	Human-led	0.566	0.879	0.111	1.31
	Agent as executor	0.748	0.904	0.077	1.07

Cost and minutes are averaged across the two agents. Source: τ^2-bench telecom trajectories.

designs the agent executes one run of length $n = n_u + n_a$; then $-\log S = \phi + p\,n + \rho\,n(n-1)/2$. The boundary-crossing hazard is $\varepsilon = h_u^{\text{shared}} - h_u^{\text{human-led}}$. We report the median and 90% interval from 200 task-cluster bootstrap replications and a likelihood-ratio test of $\rho = 0$ against the 5% critical value of χ^2_1 (3.84). The null lies on the boundary of the parameter space, so the reference distribution is conservative. Because repeated trials are highly correlated, the pooled likelihood overstates the information in the data. We therefore treat the bootstrap interval for ρ as the primary evidence and the test as corroboration. Codification effects compare $\hat{\varepsilon}$ across policies within agent. *Study 3* feeds the estimates into the model to locate each cell on the phase diagram and compute optimal designs and horizon thresholds.

Study 4 uses the public leaderboard (46 text submissions across its domains), which reports pass^k but, for most recent submissions, no trajectories we could reach. For each of the 21 text submissions on shared-control telecom we compute the per-action hazard h that solves $\exp(-h\,n_u - 0.01\,n_a) = \text{pass}^1$, averaging over the 2025 task mix with the agent-side hazard fixed at 0.01. Later submissions used a newer user simulator and task set. These hazards are therefore indicative. They cannot separate ε from p_H or identify ρ. For the 18 banking submissions evaluated between May and August 2026 we compare pass^4 with pass^1 to measure how far single-run success overstates consistent success.

4.6. **Design of Studies 5–8.** Studies 5 and 6 are new runs in a pilot phase and a main phase under a budget guard with hard caps of $2 (pilot) and $5 (main phase). The main phase supplies the results. The pilot is provenance only; we do not pool it with the main phase. *Study 5* re-runs the four designs with GPT-5.6 Luna and a GPT-4.1 mini customer, one trial per task, on 45 length-stratified tasks per design under both the prose and the codified policy (360 conversations). Eight of the tasks overlap the pilot, so we repeat every test without them. Comparisons with 2025 are indicative. *Study 6* pairs an autonomous and a collaborative agent on ten instances of Collaborative Gym's tabular-analysis task [31]. *Study 7* regresses outcomes in the real-user logs [32] on a quadratic in the human's share of actions; a positive squared term would signal a valley. The data are observational. *Study 8* uses a within-run permutation test to ask whether decisive errors in Who&When's failed runs [37] fall on hand-off steps more often than chance. It also tabulates MAST/MAD failure modes [8]. Online Appendix F gives the full procedures.

4.7. **Threats to validity.** Changing rights together with their tools and information is the construct, not a confound. The "person" in Studies 1–6 is a language model, likely more consistent than a real customer, so $\hat{\varepsilon}$ may understate human-side variation. Only Studies 1–3, 5 and 6 assign allocations; the real-user and multi-agent evidence is observational. Studies 5 and 6 are small by design, so we treat them as tests of direction and order of magnitude.

TABLE 3. Scorecard: hypotheses and the evidence bearing on them

Claim	Evidence	Verdict
H1 Unbundling valley	Shared control worst in 4 of 4 agent–policy cells, 23–57 points below the best bundled design (Study 1); 2026 agent: worst under both policies, valley test met under prose, not under codified (Study 5)	supported (2025); prose only (2026)
H2 Boundary-crossing hazard	$\hat{\varepsilon} = 0.03$–0.33, 90% interval above zero in 4 of 4 cells (Study 2); 2026: 0.33 and 0.19, both above zero (Study 5)	supported
H3 Cascades	Bootstrap interval for ρ excludes zero in 6 of 8 agent–policy–design cells (Study 2); 2026 full autonomy: about 0.05 under both policies (Study 5); with fault-count fixed effects, significant only for o4-mini (not pre-registered)	supported for o4-mini; confounded with difficulty for GPT-4.1 (prose) and GPT-5.6 Luna
H4 Codification	$\hat{\varepsilon}$ cut by a median 76% (GPT-4.1) and 58% (o4-mini) (Study 2); about 43% for the 2026 agent (Study 5)	supported
H5 Substitution	Interaction −12.3 pp (SE 2.8) (Study 1); 2026: +5.6 pp (SE 9.4) (Study 5)	supported (2025); not supported (2026)
Collaboration where the person holds knowledge	Simulated user: mean rating 0.80 with and without collaboration, 10 pairs (Study 6)	no advantage
Valley with real people	Co-Gym: inverted U, peak at a human share of 0.30; CollabSkill: no curvature (Study 7)	not supported
Hand-off hazard between agents	1.69× chance in orchestrator-led runs; 0.98× in expert teams (Study 8)	supported in one architecture

2026 values come from the main phase (45 tasks per design and policy, one trial); $\hat{\varepsilon}$ and $\hat{\rho}$ are medians of 200 task-cluster bootstrap replications. No study tests Proposition 4 directly.

Section 7.4 and Online Appendix G discuss these threats. The repository fixes the sampling rule, the budget guard and every analysis for Studies 5–7. Before the main phase we froze them by hash, together with the runtime patches the main phase needed. All 23 frozen files still matched after collection.

5. RESULTS

Table 3 summarizes the evidence by hypothesis. The controlled experiment (Studies 1–3) supports four of the five hypotheses. It supports the fifth, cascades, for o4-mini. For GPT-4.1 under the prose policy, the data cannot separate the curvature from diagnostic difficulty. A 2026 agent again did worst with split rights. Its valley test held only under the prose policy. It showed no substitution. Its cascades are confounded with difficulty in the same way (Study 5). The observational evidence is mixed. Among real users of a collaborative agent, outcomes were highest at an intermediate division of labor, the opposite of the valley we pre-specified for Hypothesis 1. Multi-agent failure logs are consistent with a hand-off hazard in orchestrator-led systems but not in expert teams, where nearly every step is a hand-off.

5.1. **Study 1: the allocation of rights.** *Shared control (the agent plans, the person executes) was the worst design in every cell.* Across the four agent–policy cells it fell 13 to 25 percentage points below full autonomy, 5 to 54 points below the human-led design and 23 to 57 points below the best bundled design (Figure 3; Hypothesis 1). It was also the most expensive design in every cell and the slowest in three of four cells and on average. A shared-control case took 2.1 minutes and $0.17 in compute on average, against 1.4 minutes and $0.08 under full autonomy: about 1.6 times the elapsed time and twice the cost. The cost figure includes the simulated customer's spend; the success gap does not depend on it.

The value of delegating execution depends on who holds the plan. Delegating execution raised success by 18.0 percentage points (SE 3.1) when the agent also planned but by only 5.6 points when the plan was specified. The interaction of −12.3 points (SE 2.8) is precisely

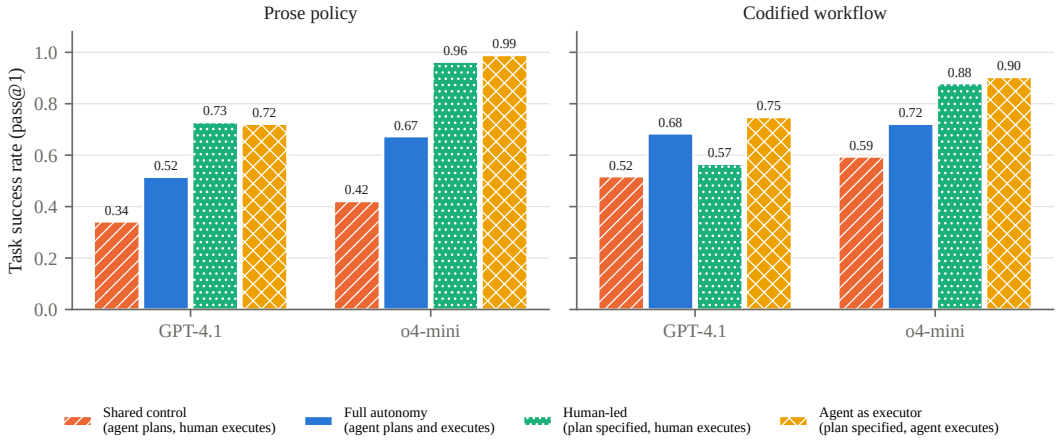


FIGURE 3. Task success by allocation of planning and execution rights, τ^2-bench telecom (456 trajectories per bar).

TABLE 4. Within-task effects of decision rights (Study 1)

	Success	Cost ($)	Seconds	Messages
Agent executes	0.180	−0.087	−47.3	−24.7
	(0.031)	(0.004)	(2.8)	(0.8)
Plan specified	0.315	−0.062	−54.7	−18.1
	(0.023)	(0.003)	(2.5)	(0.9)
Agent executes × plan specified	−0.123	0.055	34.5	8.8
	(0.028)	(0.003)	(2.9)	(0.8)
Task × agent × policy FE	yes	yes	yes	yes
Observations	7,296	7,296	7,296	7,296

Linear models; standard errors clustered by task (114 clusters) in parentheses. All coefficients significant at $p < 0.001$. The omitted category is shared control (agent plans, human executes).

estimated (Hypothesis 5). Planning retention and execution delegation are therefore substitutes (Table 4), as Proposition 3 predicts. Cost, time and messages show the same pattern: the interaction offsets between a third (messages) and three-quarters (time) of the effect of delegating execution. Agent-as-executor was the best design in three of four cells and within a point of the human-led design in the fourth. Both plan-specified designs execute the benchmark's ground-truth plan, so their success is an upper bound for human-held planning. A persona check is consistent with a structural interface hazard but is not a strong test (Online Appendix H).

5.2. **Study 2: estimating the two hazards.**

Boundary-crossing hazard. *Each instruction from the agent to the customer carried a substantial risk that we can identify separately.* Under shared control success fell from 0.86 with no customer actions to 0.06 with seven or more. Under the human-led design, on the same tasks, it stayed above 0.65 up to six actions (Figure 4, right). The per-action hazard of customer-held steps was 0.15 to 0.34 under shared control and 0.01 to 0.12 with the plan specified (Table 5). Their difference, $\hat{\varepsilon}$, ranges from 0.03 (GPT-4.1, codified) to 0.33 (o4-mini, prose). Its 90% bootstrap interval lies above zero in all four cells (Hypothesis 2), though only barely for GPT-4.1 under the codified policy (lower bound 0.001). In the worst cell, an instruction to the simulated customer to perform one action carried a one-in-three chance of derailing

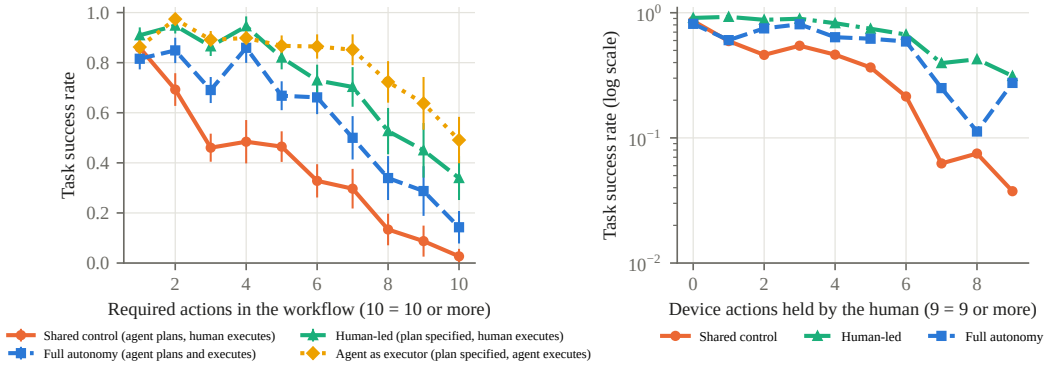


FIGURE 4. Left: success by number of required actions (95% intervals). Unsupervised designs decay slowly at first and then sharply. Right: success by number of actions held by the human, log scale; geometric decay appears as a straight line. The steeper slope under shared control is the boundary-crossing hazard.

TABLE 5. Structural estimates (Study 2), with 90% task-cluster bootstrap intervals

Policy	Agent	Hazard per human action		$\hat{\varepsilon}$	$\hat{\rho}$ (full autonomy)
		Shared control	Human-led		
Prose	GPT-4.1	0.215 [0.146, 0.293]	0.074 [0.045, 0.107]	0.134 [0.078, 0.203]	0.023 [0.014, 0.032]
	o4-mini	0.338 [0.270, 0.398]	0.011 [0.006, 0.017]	0.329 [0.260, 0.386]	0.040 [0.024, 0.055]
Codified	GPT-4.1	0.152 [0.112, 0.208]	0.123 [0.084, 0.165]	0.032 [0.001, 0.070]	0.000 [0.000, 0.015]
	o4-mini	0.182 [0.140, 0.216]	0.040 [0.029, 0.050]	0.139 [0.102, 0.174]	0.024 [0.016, 0.034]

$\hat{\varepsilon}$ is the bootstrap median of the difference between the two hazards. $\hat{\rho}$ is the cascade intensity in the single unsupervised run of the full-autonomy design. Estimated planning failure under full autonomy is 0.44 for GPT-4.1 with the prose policy and below 0.08 elsewhere.

the case. These figures are lower bounds on the full cost of the interface, because a specified plan still reaches the customer through the agent's relay.

Cascades. *Unsupervised runs degraded faster than independent errors allow. The curvature survives difficulty controls for o4-mini but is confounded with diagnostic difficulty for GPT-4.1 under the prose policy.* Under full autonomy, success stayed between 0.69 and 0.86 for workflows of up to four actions and then fell steeply, to 0.29 at nine actions and 0.08 at ten (Figure 4, left). The task-cluster bootstrap interval for $\hat{\rho}$, our primary evidence, excludes zero in six of eight agent–policy–design cells (Hypothesis 3). The likelihood-ratio test rejects $\rho = 0$ in the same six cells (statistics 4.8 to 36.9). Under full autonomy the exception is GPT-4.1 under the codified policy, whose errors were frequent but did not compound. Proposition 1 applies to that cell.

Longer tasks are also harder to diagnose. The executor design brackets this confound: supplying the plan removes both the harder diagnosis and the compounding of commitments to it. In the three cells where full autonomy compounds, the executor's $\hat{\rho}$ is lower: about 0.009 against 0.023 and 0.024 in two cells and 0.0005 (not significant) against 0.040 for o4-mini under the prose policy. Only for GPT-4.1 under the codified policy does the executor compound modestly (0.016, interval 0.005–0.023) where full autonomy does not. This fits cascades that arise from building on an early, possibly wrong, diagnosis.

Part of the full-autonomy curvature may still reflect difficulty. We test this with controls that we did not pre-register (Online Appendix I). Under full autonomy, adding a linear fault count or the persona leaves the bootstrap interval for $\hat{\rho}$ above zero in every cell with cascades. With fault-count fixed effects, the most flexible control, curvature survives only for o4-mini

(prose 0.037, LR 17.1; codified 0.023, LR 6.4), not for GPT-4.1 under the prose policy (0.007, LR 1.2). The executor's curvature vanishes. Net of task fixed effects, full autonomy's curvature exceeds the executor's in all three 2025 cells with cascades (LR 4.0 to 27.5). That contrast places the curvature in agent-held planning but cannot separate compounding from diagnostic difficulty.

Codification. *Expressing the policy as an explicit workflow made the interface far less lossy.* Codification reduced the boundary-crossing hazard from 0.134 to 0.032 for GPT-4.1 and from 0.329 to 0.139 for o4-mini, median reductions of 76% (90% interval 48–100%) and 58% (48–68%) (Hypothesis 4). Codification is thus an interface technology, the only lever in our data that lowered what is lost at the boundary without changing who holds which right. It also removed GPT-4.1's cascades under full autonomy.

5.3. **Study 3: calibrated optimal designs.** *Fed back into the model, the same benchmark recommends very different policies for different agent–policy combinations.* On the phase diagram (Figure 2), GPT-4.1 under the codified policy sits at low ε and zero ρ, so full autonomy is optimal at every workflow length we compute. The other three cells sit where human-led work dominates for workflows of up to 12 steps. For GPT-4.1 under the prose policy, supervised autonomy with $a^* = 0.83$ becomes optimal at 24 steps. The valley of Proposition 3 is present throughout: in every cell and at every workflow length we compute (4 to 24 steps), the first unit of autonomy lowers value relative to the human-led design. For the three cells with $\hat{\rho} > 0$, the horizon threshold of Proposition 2 restricts full autonomy to at most four or five steps when the remaining touchpoints are shared-control actions. When they are well-specified review gates, the limit is one to three steps. In those cells the optimal autonomy level among agent-led designs for a 12-step workflow is 0.83 with shared-control touchpoints and 0.50 to 0.75 with specified gates.

The human-led corner inherits the benchmark's ground-truth plan, the best case for human planning. Labor costs are set at reference values ($c_H = 0.01R$; Online Appendix B). Neither choice affects the qualitative conclusions.

5.4. **Study 4: the 2025–2026 frontier.** *The interface hazard in structured work has fallen; knowledge-intensive work has not followed.* For each text submission on shared-control telecom we computed the per-action hazard on customer-held steps that reproduces its reported pass^1 on the 2025 task mix. The median implied hazard fell from 0.26 among the four models of the June 2025 release to 0.025 among models evaluated in 2026. The user simulator and the task set also changed, so these values are indicative. In the knowledge-intensive banking domain, the 18 submissions evaluated between May and August 2026 averaged a pass^1 of 0.34; pass^4 averaged 49% of pass^1. Online Appendix J holds the simulator fixed and details banking (Figure S2).

5.5. **Studies 5 and 6: a 2026 agent and a second kind of work.** *With a 2026 agent, splitting rights was again the worst design, but the valley test held only under the prose policy.* GPT-5.6 Luna ran once per design on 45 length-stratified telecom tasks under each policy, 360 conversations in all (Table S3 in Online Appendix K). Under the prose policy shared control succeeded in 35.6% of tasks, against 60.0% for full autonomy, 80.0% for the human-led design and 100% for the executor. Under the codified policy the rates were 46.7%, 55.6%, 75.6% and 100%. The pre-specified valley test of Hypothesis 1 is met under the prose policy: shared control trailed full autonomy by 24.4 points ($z = 3.39$, one-sided $p < 0.001$) and the human-led design by 44.4 points ($z = 4.78$). It is not met under the codified policy. There the gap to the human-led design was 28.9 points ($z = 3.10$, $p < 0.001$), but the gap to full autonomy was only 8.9 points ($z = 1.16$, $p = 0.12$). Without the eight tasks that overlap the pilot, both conclusions stand.

Corollary 1, checked after the fact, accounts for the codified result: a shallower valley against the human-led design, not its removal. We evaluated Corollary 1 at the estimates of all six agent–policy cells, a check we did not pre-register (Online Appendix I). The critical

hazard ε^* is below 0.02 (negative for GPT-4.1 at the task mix). Every codified point estimate of ε exceeds it (0.03 to 0.20; bootstrap medians 0.03 to 0.19). Removing the valley would require cutting ε by more than 98%; codification cut the point estimates by 41–79% (bootstrap medians 43–76%). Lower ε explains only part of the modeled shallowing. A weaker human-led corner under codification explains the rest. The data match for all three agents: under the codified policy shared control still trailed the human-led design, by 5 to 29 points (39 to 54 under prose). Modeled and observed gaps rank alike (Spearman 0.94 against the human-led design and 0.83 against full autonomy). This does not conflict with the failed valley test. The corollary concerns the human-led corner ($V(1) < V(0)$); the pre-specified test of Hypothesis 1 also requires shared control to lie below full autonomy. Under the codified policy that 8.9-point gap was not significant.

The hazards repeat the 2025 pattern. Under the prose policy $\hat{\varepsilon} = 0.325$ [0.216, 0.514]; codification lowered it to 0.187 [0.098, 0.284], by about 43%. Both intervals lie above zero. Under full autonomy $\hat{\rho}$ was 0.046 [0.030, 0.062] (prose) and 0.052 [0.035, 0.080] (codified), above the 0.023–0.040 of the 2025 cells with cascades. The executor never failed. The agent's failures under full autonomy are thus associated with holding the plan. As in Study 2, however, the cascade term cannot separate compounding from diagnoses that grow harder with length. With fault-count fixed effects $\hat{\rho}$ stays positive (0.031 prose, 0.030 codified), but its intervals reach zero (LR 3.2 and 1.3; Online Appendix I).

Hypothesis 5 did not hold for this agent. Delegating execution raised success by 16.7 points (SE 6.6) and specifying the plan by 36.7 points (SE 8.7). The interaction was +5.6 points (SE 9.4, $p = 0.56$), against −12.3 in 2025. In our reading the relay stayed costly even with a specified plan: the human-led design lost a fifth or more of the cases that the executor solved. The executor's 100% holds for these tasks only. Because it rests on the benchmark's ground-truth plan, it is an upper bound for human-held planning.

Where a simulated user held knowledge the agent needed, collaboration did not raise scores. In Study 6 the autonomous and the collaborative agent each reached a mean rating of 0.80 on ten paired Collaborative Gym analysis tasks [31] (paired difference 0.00, SD 0.47). Both completed every task. The simulated user took 11% of task actions. Ten pairs cannot rule out a modest effect in either direction. They give no sign, however, that the knowledge channel we conjecture from the real users of Study 7 operates when the knowledge holder is a language model (Online Appendix K).

5.6. **Beyond the laboratory: real users and multi-agent systems.** *Among real users of a collaborative agent, outcomes peaked at an intermediate division of labor, the opposite of the valley we pre-specified.* The public release of Collaborative Gym real-user logs holds 228 sessions (the original paper analyzed 150 trajectories; [31]). Of these, 188 had usable logs and ratings. The 1–5 outcome rating rose and then fell with the human's share of task actions (linear term +2.70, squared term −4.50, $p = 0.001$). It peaked at a human share of 0.30, about 0.4 points above a zero share. Hand-offs per action were unrelated to the rating ($b = -0.17$, $p = 0.68$). The decline beyond the peak rests on very few sessions: only six had a human share above 60%.

We do not read this as support for Hypothesis 1. The valley's premise fails here: in Co-Gym's travel-planning, writing and analysis tasks the user holds preferences and context the agent lacks, which moderate participation may supply. Users also chose their shares and rated the outcome themselves (Online Appendix L). The valley warning concerns splits of execution; whether consultation pays remains a conjecture (Section 7).

Professional workers showed neither a valley nor a peak. In CollabSkill [32], 262 of 386 sessions had usable logs and grades. The graded score showed no significant curvature in the human share (squared term −12.4, $p = 0.59$) and no relation to hand-offs per action ($b = -4.0$, $p = 0.53$).

In an orchestrator-led system, decisive errors cluster at hand-offs; in expert teams, they do not. In the orchestrator-led Magentic-One runs of Who&When [37] (58 failed runs), the

decisive error fell 45 times on a step at which control had just passed to a different agent, against 26.6 expected if errors ignored hand-offs: 1.69 times the chance share (within-run permutation test, $p < 0.001$). In the CaptainAgent expert teams (126 failed runs), where about five steps in six are hand-offs, the ratio was 0.98 ($p = 0.79$), a weakly informative null rather than a contrary one. Hand-off losses thus appear where a planner delegates to executors, not where every step is a hand-off. That hand-off is a natural place for checks. In the MAST/MAD traces [8], 1,638 of 1,642 had usable labels. The shares of traces with at least one specification failure (0.58), inter-agent misalignment (0.53) or verification failure (0.55) were similar, so none of the model's three failure channels is minor. A firm should not assume that a verifier that is just another agent in the chain is a reliable touchpoint. Online Appendix L reports the remaining detail.

6. From Theory to Practice: Workflows and Algorithms

Firms do not set an "autonomy level". They decide which tools an agent may call, which steps a person must perform or approve, who writes the runbook and where someone looks at the work before it moves on. Each choice allocates planning, execution or observation rights. We now turn the theory into procedures for practitioners. They are a touchpoint-placement algorithm, an audit of ϕ, ε, ρ and κ from a firm's own logs, two control algorithms and two deployment workflows. The script `07_practice.py` produces its new numbers.

6.1. Placing touchpoints with heterogeneous steps.

In real workflows some steps only one party can take and some are riskier than others. Algorithm S1 (Online Appendix M) is an exact dynamic program that finds the value-maximizing allocation of execution rights for any sequence of steps under an agent-held plan. On the longest pilot task (eleven actions, nine on the phone) with the hazards of o4-mini under the prose policy, it hands the customer the fourth and eighth actions. Modeled success rises from 0.06 under full autonomy to 0.27. It is 0.02 when the customer performs all nine device actions and 0.97 with a specified plan; the latter is an upper bound for human-held planning (Online Appendix N). Who holds the plan is the first decision; placing touchpoints is the second.

6.2. A decision-rights audit from agent telemetry.

The audit (Algorithm S2 in Online Appendix O) is Study 2 run inside a firm, with one change: the firm must create the variation in allocations deliberately. It assigns a share of cases at random, within strata of workflow length, to a design that differs from production in one right. It fits the likelihoods of Study 2 to the requestor of each logged action and takes $\hat{\varepsilon}$ as the difference between human-step hazards under agent-held and specified plans. It returns a design, a horizon threshold $\bar{n}$ and the investment with the largest value per dollar, or *insufficient data* with the sample still needed. A few dozen cases per design reveal a large valley. Locating a workflow on the phase diagram takes about a hundred or more per design (Online Appendix O.2).

6.3. Deployment workflows and control procedures.

A deployment must also decide when the evidence justifies moving the planning right and when a person should look at a run in progress. Algorithms 1 and 2 make these two decisions. Figures 5 and 6 place all four procedures in a firm's workflows. They are procedures the model implies. We have not tested them in deployment. Their inputs are the estimates and the audit of Sections 5 and 6. The replication materials hold a reference implementation. Runbooks need a test for gaps; our plan-specified designs had a flawless plan.

Algorithm 1 decides who holds the planning right for a workflow class w by comparing the failure probability of a crossing step, $q = p_H + \varepsilon$, with the threshold of Corollary 1. Crossings are few at first, so it uses a posterior for q and acts only after $m_{\min}$ crossings and ℓ consecutive windows above confidence τ. It then picks the design with the highest V at $\hat{\varepsilon}_w$, which a large $\hat{\varepsilon}_w$ steers toward bundled rights. It always reports the codification cut c_{req} that removes the dip. When $\varepsilon^*_w = 0$, codification cannot make sharing viable.

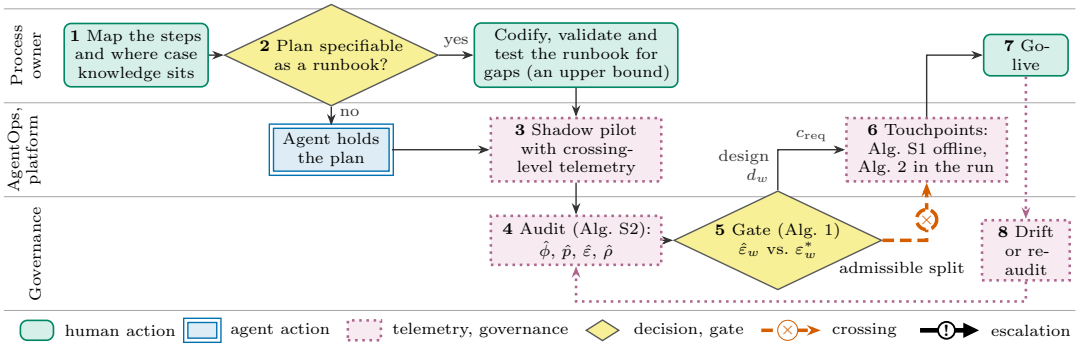


FIGURE 5. Deploying an agent on a process: the allocation of rights before and at go-live, in numbered steps. Color online; in print, shapes and line styles carry the same coding.

Algorithm 1 Hazard-gated reallocation of decision rights for a workflow class w (between cases)

Input: closed cases in w (per step: planner, executor, outcome, crossing); estimates $\hat{p}_H, \hat{p}_A, \hat{\phi}_H, \hat{\phi}_A, \hat{\rho}$, n_w; c_H, c_A, R; prior $\text{Beta}(\alpha_0, \beta_0)$ on $q = p_H + \varepsilon$; confidence τ; minimum crossings $m_{\min}$; hysteresis ℓ; design d_w

Output: design d_w, codification cut c_{req}, decision log

[T] $\varepsilon^*_w \leftarrow$ equation (6) at the estimates; $\varepsilon^*_w \leftarrow 0$ if $a + \gamma_0 \le 0$; $\varepsilon^*_w \leftarrow \infty$ if $\Delta c \ge S_0$ ▷ Corollary 1

[T] $t, f \leftarrow$ crossing steps, failures; $q \sim \text{Beta}(\alpha_0 + f, \beta_0 + t - f)$; $\hat{\varepsilon}_w \leftarrow \max\{0, \text{E}[q] - \hat{p}_H\}$; $P_w \leftarrow \Pr(q > \hat{p}_H + \varepsilon^*_w)$

if $t \ge m_{\min}$ and $P_w > \tau$ in ℓ consecutive review windows **then** ▷ reallocate

    [H] $d_w \leftarrow$ design with the highest V (eq. 2, at $\hat{\varepsilon}_w$) among human-led, agent as executor (if a runbook is validated), full autonomy and agent-led with touchpoints (Algorithms S1 and 2)

else if $t \ge m_{\min}$ and $P_w < 1 - \tau$ **then**

    [H] shared control is admissible; $d_w \leftarrow$ design with the highest V, shared control included

else

    [T] keep d_w; collect more crossings

end if

[T] $c_{\text{req}} \leftarrow 0$ if $\hat{\varepsilon}_w \le \varepsilon^*_w$, else $1 - \varepsilon^*_w / \hat{\varepsilon}_w$; log d_w, its inputs, P_w and the reason ▷ cut that removes the dip

Algorithm 2 works inside a run. At each agent step it weighs two quantities. One is the cascade loss that a touchpoint would avoid before the next scheduled one. The other is the touchpoint's labor and, under an agent-held plan, its boundary hazard.[1] Only a vigilant touchpoint resets the run, so canary probes in the review packets measure κ. Canaries enter only the reviewer's packet, never the live system or the customer's case. When $\hat{\kappa}$ falls below a floor, the rule buys observability instead of review, as Proposition 4 implies.

6.4. **Where the tools stop.** These tools inherit the model's scope and the limits of its estimates (Online Appendix O). The model has no channel through which a person improves the plan, so the tools charge each consultation as a boundary crossing. Firms should not use them to remove people whose contribution is knowledge. The tools read $\hat{\rho}$ as compounding and may overstate what touchpoints buy. The worked example and the sample sizes come

[1]Making step i a touchpoint changes its hazard from $p_A + \rho u$ to $p_H + \varepsilon \mathbf{1}[\pi = A]$ and moves the next L agent steps from depths $u+1, \ldots, u+L$ to $0, \ldots, L-1$ if it is vigilant or to $u, \ldots, u+L-1$ if not. To first order, $\Delta \log S \approx \rho[u + L(1 + \kappa u)] - (p_H - p_A) - \varepsilon \mathbf{1}[\pi = A]$; $\Delta V \approx R\hat{S}\, \Delta \log S - (c_H - c_A)$ gives G and C. At $\kappa = 1$ and $u = L = (n-1)/2$ this is equation (3).

Algorithm 2 Cascade-aware adaptive checkpointing with vigilance probes (within a run)

Input: estimates $\hat{p}_A, \hat{p}_H, \hat{\varepsilon}, \hat{\rho}, \hat{\phi}_A$; planner π; costs c_H, c_A; value R; scheduled touchpoints (Algorithm S1); canary rate q_c; vigilance floor $\kappa_{\min}$ after at least m_c canaries; posterior $\mathrm{Beta}(a_\kappa, b_\kappa)$ on κ

Output: touchpoint positions, trace of $\hat{\kappa}$, escalation flags

$u \leftarrow 0$ ▷ steps since the last effective touchpoint

for each step $i = 1, \ldots, n$ **do**

  [T] $\hat{\kappa} \leftarrow a_\kappa/(a_\kappa + b_\kappa)$; $L \leftarrow$ agent steps to the next scheduled touchpoint

  [T] $\hat{S} \leftarrow$ modeled success of the case under the current schedule (eq. 1)

  [T] $G \leftarrow R\hat{S}\hat{\rho}\,[u + L(1 + \hat{\kappa}u)]$; $C \leftarrow (c_H - c_A) + R\hat{S}\,[\hat{p}_H - \hat{p}_A + \hat{\varepsilon}\,\mathbf{1}[\pi = A]]$

  if i is scheduled, $G > C$ or an anomaly (tool error, plan revision, state mismatch) **then**

    [H] a person executes or reviews step i; with probability q_c the review packet holds a canary

    [T] canary caught: $a_\kappa \leftarrow a_\kappa + 1$, $u \leftarrow 0$; missed: $b_\kappa \leftarrow b_\kappa + 1$, flag escalation; no canary: $u \leftarrow (1 - \hat{\kappa})u$

  else

    [A] the agent executes step i; $u \leftarrow u + 1$

  end if

  [T] **if** $\geq m_c$ canaries and $\hat{\kappa} < \kappa_{\min}$: flag; give the reviewer snapshots and replay logs or cap the run length u at the horizon threshold $\bar{n}$ (eq. 4) with $\tilde{\rho} = \hat{\rho}(2 - \hat{\kappa})/\hat{\kappa}$

end for

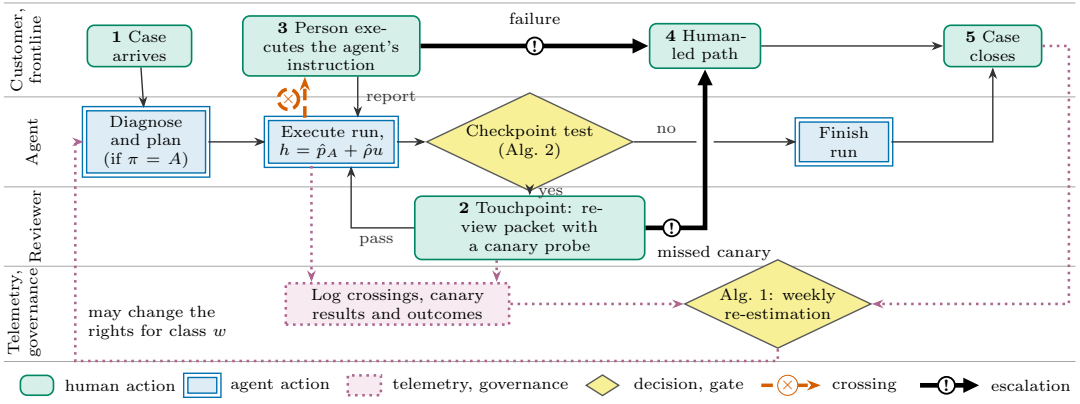


FIGURE 6. Live operations: one case through hand-offs, checkpoints and escalation. A failure or a missed canary sends the case to the human-led path; closed cases feed Algorithm 1, which can change the rights for the class. Color online; in print, shapes and line styles carry the same coding.

from one benchmark domain with simulated customers. We have not validated the audit between agents.

7. Discussion

The evidence is uneven, so we first sort it by strength. The strongest comes from a laboratory that held the task, the environment and the simulated customer fixed and varied only the allocation of rights (Studies 1–3). The 2026 replication (Study 5) and the collaboration experiment (Study 6) also assign designs, but they are small. The frontier evidence is indicative (Study 4). The real-user and multi-agent evidence is observational (Studies 7 and 8). Read this way, the valley and the boundary-crossing hazard rest on causal evidence for all three agents, though the 2026 valley test held only under the prose policy. Cascades rest on it for o4-mini alone. The knowledge channel rests on observational evidence; the one experiment

(Study 6) did not find it. The subsections below draw out its meaning for theory, managers and research, then state its limits.

7.1. **Theoretical contributions.** *Autonomy as a bundle of rights.* The paper replaces the scalar "level of autonomy" with an allocation of rights over the steps of a workflow: to plan and to execute, which carries the right to observe. For the 2025 agents an execution right was worth about three times as much when the agent also held the plan, a finding that a scalar framework cannot even state. For the 2026 agent the interaction was not distinguishable from zero (+5.6 points, SE 9.4), so the 2025 complementarity did not replicate in this sample. Our data cannot say whether it differs by agent. This extends the delegation view of IS use [4] from tasks to rights within a task. It gives autonomy, one of the facets that make AI hard to manage [6], a measurable unit. As in multitask agency, the value of assigning one element depends on what else is grouped with it [15]. Here, though, the interaction runs through information and error rather than incentives. Consistent with Aghion and Tirole [1], an agent with formal authority over a plan for a device it cannot see has little real authority over the outcome.

Two hazards in the structure of the work. Research on human–AI collaboration has mostly explained failures of joint work by human limitations: aversion, overreliance, complacency, miscalibrated trust [10, 12, 21, 25]. Our model locates two failure mechanisms in the structure of the work. Both operate with competent, attentive people. The boundary-crossing hazard ε belongs to the interface between planner and executor. It was substantial in most laboratory cells and for the 2026 agent, shrank sharply under the codified policy and showed no detectable dependence on how difficult the simulated customer was. The cascade intensity ρ belongs to the unsupervised run. Our evidence for it is weaker: for GPT-4.1 under the prose policy and for the 2026 agent, the data cannot separate it from diagnostic difficulty. Its micro-foundation is the agent's loop, in which each step conditions on the agent's own earlier interpretation [11, 36]. It resembles a latent error that later actions build on [30] and spreads as failures do in tightly coupled systems [27]. The remedies differ. Codification and bundled rights address the first hazard; touchpoints, checkpoints and observability address the second. Where the problem is boundary crossings, approval gates make things worse, because each gate under an agent-held plan is another crossing.

When augmentation is optimal. The automation–augmentation debate holds that the two are interdependent over time without saying when a mixed allocation beats either pole [29]. Propositions 1 and 2 give a condition for one case. When the person holds only execution rights the agent could hold, an interior allocation is optimal only if errors compound within unsupervised runs. Without compounding, the efficient move is to a corner, which may be abrupt rather than gradual. Our data establish compounding net of difficulty only for o4-mini.

A knowledge channel: a conjecture with mixed evidence. The model gives the person only execution, so it cannot produce an advantage for a moderate human share. Among Collaborative Gym's real users (Study 7), ratings rose with the person's share of actions to a peak near 30%. This evidence is observational (Section 7.4). When we assigned collaboration (Study 6), a simulated user who held the needed knowledge gave no advantage (0.80 in both conditions, ten pairs). That user was a language model taking about a tenth of the actions, so the result neither confirms the channel nor rules it out for real people. Among CollabSkill's professional workers scores were flat in the human share. We therefore offer the knowledge channel as a conjecture and a research direction, not a finding. The party holding specific knowledge [16] would hold a right to inform the plan, lowering ϕ but crossing the boundary like any other exchange. Organization theory anticipates the distinction between contributing to the protocols that govern action and selecting the actions themselves [23]. Such input should pay when the task's uncertainty exceeds what the planner can process alone [13]. Suppose the channel were formalized and then confirmed by randomizing consultation. The valley and an interior optimum would then be two cases of one theory. Periodic touchpoints would contain cascades; consultation on the plan would bring in knowledge.

Accountability as a control parameter. Proposition 4 links who answers for an agent's failure to how much oversight is worth. Lower vigilance raises the effective cascade intensity by the factor $(2-\kappa)/\kappa$, which implies an exchange rate between reviewer accountability and the observability of runs. It is the least-tested part of the theory, because our data do not vary vigilance. Verifier steps in multi-agent runs were no less error-prone than other steps. That is consistent with low vigilance, but it comes from failed runs only.

7.2. **Managerial implications.** Each rule below rests on a result of Section 5 and inherits its caveats.

Keep execution rights bundled. One design has the agent diagnose a case and tell a person what to do. It was the worst in every laboratory cell and again for a 2026 agent. This is the paper's best-supported finding. Handing the agent a specified plan did best in three of four laboratory cells and in both 2026 cells, but that plan was the benchmark's ground truth. Its advantage, including the 2026 executor's 100%, is an upper bound for what a firm's own runbook will deliver.

Do not automate away knowledge on the strength of the valley. The valley concerns split execution. That people who hold knowledge the agent lacks add value is a conjecture. It counsels caution before removing them; it does not show that consultation pays.

Bound unsupervised runs where errors compound. Where $\hat{\rho}$ was non-zero, full autonomy paid only for runs of four or five steps with customer touchpoints and one to three with specified review gates. These thresholds treat the curvature as compounding, a reading that survives difficulty controls only for o4-mini; elsewhere touchpoints may buy less. They hold for these agents in this domain at reference labor costs; a firm must estimate its own. A redesign that lengthens an agent's run should reopen the decision.

Codify before delegating. Codification was the only lever in our data that reduced the interface hazard without changing who held which right. Under the codified policy the valley was shallower but persisted: shared control still trailed the human-led design by 5 to 29 points. For the 2026 agent, though, its gap to full autonomy was not significant. Codification also lowered success in three of eight design cells (by 8–16 points, all with a specified plan). A firm should therefore judge it by its effect on $\hat{\varepsilon}$ in its own workflow.

Treat observability and accountability as substitutes. This rule follows from the theory, not from our data; firms should check it. A firm that cannot keep reviewers vigilant may buy back the same effective cascade intensity through tracing, checkpointing and rollback. It should measure vigilance before relying on review.

Read capability gains by workflow. On indicative leaderboard estimates, the interface hazard in structured work fell by about an order of magnitude in a year. Knowledge-intensive banking lagged far behind. A firm that sets its autonomy policy by headline gains risks over-delegating knowledge-intensive work.

Run the audit as a field experiment, with realistic expectations. Only a large valley shows up quickly; locating a workflow takes far more cases. Because the audit reallocates decision rights, it belongs with the workflow's owner, working with the platform team.

7.3. **Implications for research and for building agents.** The natural next study moves the laboratory into a firm. It would randomize planning and execution rights across matched cases in a production workflow, stratified by workflow length and only among designs the firm is prepared to run. It would estimate ε and ρ with real people; the online experiment described under Limitations is a first step. Seeding known faults upstream of reviewers would estimate κ. Varying accountability would test whether it moves κ as Proposition 4 requires. The knowledge channel should be tested by randomizing consultation rather than observing it, in settings where the human holds information the agent needs [31, 32]. Boundaries between agents need a model of the orchestrator's planning right and complete traces, not only failed ones. For those who build agents, observation should travel with action: an agent that plans should hold the tools to act on, and observe, the state it plans for. Benchmarks built to rank

agents can also serve as organizational laboratories [5] if they report pass^k [35] with cost [17] by allocation of rights. Online Appendix Q develops these points.

7.4. **Limitations.** We list the limitations in order of how much they could change our conclusions.

A benchmark is not a firm. In Studies 1–5 the person is a language model simulating a customer and is likely more consistent than real customers. Our $\hat{\varepsilon}$ may therefore understate the variation real people add. The paper's causal evidence remains simulator-based. We have designed, but not run, an online experiment with real participants. They carry out the device steps on a simulated phone. The planning right is randomized between participants (the agent's own plan or a specified plan that the agent relays). The experiment targets about 280 analyzable participants. It is powered for ε of 0.10 and, at low human hazards, 0.05. We will pre-register it before data collection (protocol in Online Appendix Q). We report no results from it. The null persona effect leaves room for moderate effects. The laboratory evidence also comes from one structured domain; Study 4's banking results suggest that knowledge-intensive work behaves differently.

The specified plan is a best case. In the human-led and executor designs the plan is the benchmark's ground truth, the equivalent of a flawless runbook. Their advantage, including the 2026 executor's 100%, is therefore an upper bound for human-held planning; real planning failure will be positive. The repository specifies an imperfect-runbook test. It reruns both designs on Study 5's 45 tasks with one plan step dropped or swapped (expected cost about $0.70). We have not run it (Online Appendix F).

The cascade parameter is only partly identified. Longer tasks combine more faults, so curvature under full autonomy may partly reflect diagnostic difficulty. With fault-count fixed effects it survives only for o4-mini; for GPT-4.1 under the prose policy and for GPT-5.6 Luna these data cannot separate cascades from difficulty. The executor contrast locates the curvature in agent-held planning but does not remove the confound. These checks, like the evaluation of Corollary 1, were not pre-registered.

The 2026 evidence is narrower than the 2025 laboratory. The leaderboard reports aggregates under a simulator and task set that changed over time. It cannot separate ε from the customer's own error rate or identify ρ. Study 5 uses one 2026 agent, one trial per cell and 45 tasks per design and policy. It can establish the direction of the results, not their magnitude. Its executor's 100% success holds only for these tasks. Its simulated customer, step limit and task sample differ from 2025, so comparisons across years are indicative (Online Appendix F). The codified domain and Collaborative Gym ran only after runtime patches, which we froze with the analysis before collection; the codified-domain smoke test used a mock model. Study 6 has ten pairs. Its grader and simulated user run on GPT-4.1 mini rather than Collaborative Gym's original GPT-4o, so its absolute scores are not comparable with published results. We reran two of its sessions after interruptions. A host interruption makes elapsed times in both studies unreliable.

The real-user and multi-agent evidence is observational. In Study 7 the division of labor was self-selected, the decline beyond the peak rests on six sessions and the outcome is the users' own rating. In Study 8, Who&When contains only failed runs; an LLM judge produced MAST's labels.

Costs are partial. Compute cost omits the labor of a real customer or employee, which in a firm is likely the larger cost of a split design.

8. Conclusion

Whether agentic AI creates or destroys organizational value depends on how a firm divides decision rights between the agent and the people around it, not only on the agent's capability. Autonomy is not a dial but an allocation of the rights to plan and to execute, with observation bound to execution. The value of each right depends on who holds the others.

The finding that travels furthest is the valley. Splitting planning from execution cost 23 to 57 percentage points against the best bundled design in a controlled laboratory and was again worst for a 2026 agent. For the 2026 agent under the codified policy the valley was shallower: shared control still trailed the human-led design, but the pre-specified test against full autonomy failed. Other claims are narrower. Cascades are separable from harder diagnoses only for o4-mini, the substitution between rights did not replicate in 2026 and the executor's advantage is an upper bound, because it rests on a flawless plan. Whether consulting people who know what the agent does not pays remains a conjecture: real users hint at it, but our experiments do not confirm it.

Managers should decide rights, not levels. Keep execution with whoever holds the plan, or give the person a plan worth executing. Bound unsupervised runs where errors compound. Codify before delegating. Audit the allocation rather than assume it. Move the planning right only on evidence.

The autonomy paradox is real, but it is not a mystery. It has a structure (two hazards, a valley and possibly a boundary) that firms can measure and manage.

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